Mise en place 'Sélection Génomique' par G-BLUP, 1iere session. INP - AgroToulouse

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```
#############################################################################################
###                                Session Genomic Selection                              ###
###  Case study #1: Prediction of yield depending on Environments(BLUE)and genotypes(BLUP)- ###
###  Dataset created from experimental design described in Roumet et al., 2012, New Phytol  ###
###                                        ----                                           ###
###       R Script edited by Pr Laurent Gentzbittel & Cécile Ben, INP-ENSAT                 ###
#############################################################################################
```

1 Preparing workspace

```
######### PREPARATION OF THE WORKING INTERFACE IN R ##################################################
### I. Set working directory
#On RStudio: tab 'Session'-> Set Working Directory -> Choose Directory.
#Choose the directory containing the datafile and the associated R script.

### II. Installation R packages needed for the analysis on RStudio:
#Click on the 'Packages' tab in the bottom-right window of R Studio interface->'Install Packages'
#Comment #1: R package installation requires a connection to internet
#Comment #2: Once packages have been installed, no need to re-install them again when you close-open ag

### III. Initialisation of the working space:clean workspace, clean memory
# To erase all graphs
graphics.off()
# To erase objects from the working space - Clean up of the memory
rm(list = ls())
```

```
# garbage collector. To use to free memory
gc()
```

```
##          used (Mb) gc trigger (Mb) max used (Mb)
## Ncells 485353 26.0     940480 50.3   750400 40.1
## Vcells 917664  7.1    1650153 12.6  1073553  8.2
```

```
################################################################################################################################

## Loading of the R packages needed for the analysis.
library(rrBLUP)
library(lme4)
```

```
## Loading required package: Matrix
```

```
## Loading required package: methods
```

2 Loading data.

```
####################
# Data import in R
####################
#########
## EX. 1: Computation of BLUEs & BLUPs to predict Yield using environnement and Genotype WITHOUT MARKER
#########

sunflower <- read.table("GS1_SunflowerWeed.csv", sep=";", header=TRUE, dec=".")
sunflower #Dataset 'inspired' from Roumet et al., New Phytol, 2012
```

```
##             ENV       Geno YEAR  YLD
## 1         Odars NK Countri 2007 1489
## 2         Odars    PR64H41 2005 1526
## 3       Verfeil NK Countri 2007 1473
## 4       Verfeil    PR64H41 2005 1535
## 5  Villefranche    PR64H41 2007 1486
## 6  Villefranche     Atomic 2005 1426
## 7  Villefranche   Santiago 2003 1578
## 8   Fourquevaux NK Countri 2007 1483
## 9   Fourquevaux      Boogy 2005 1562
## 10  Fourquevaux    Aurasol 2003 1511
## 11        Balma    PR64H41 2007 1463
## 12        Balma    PR64H41 2005 1496
```

```
str(sunflower)
```

```
## 'data.frame':    12 obs. of  4 variables:
##  $ ENV : Factor w/ 5 levels "Balma","Fourquevaux",..: 3 3 4 4 5 5 5 2 2 2 ...
##  $ Geno: Factor w/ 6 levels "Atomic","Aurasol",..: 4 5 4 5 5 1 6 4 3 2 ...
##  $ YEAR: int  2007 2005 2007 2005 2007 2005 2003 2007 2005 2003 ...
##  $ YLD : int  1489 1526 1473 1535 1486 1426 1578 1483 1562 1511 ...
```

```
#5 genotypes of cultivated sunflowers, 6 environments, 3 years,
##5 genotypes = markers in real GS datasets

sunflower$YEAR<-as.factor(sunflower$YEAR)
str(sunflower)
```

```
## 'data.frame':    12 obs. of  4 variables:
##  $ ENV : Factor w/ 5 levels "Balma","Fourquevaux",..: 3 3 4 4 5 5 5 2 2 2 ...
##  $ Geno: Factor w/ 6 levels "Atomic","Aurasol",..: 4 5 4 5 5 1 6 4 3 2 ...
##  $ YEAR: Factor w/ 3 levels "2003","2005",..: 3 2 3 2 3 2 1 3 2 1 ...
##  $ YLD : int  1489 1526 1473 1535 1486 1426 1578 1483 1562 1511 ...
```

```
##Vector of observations-------------------------------------------------------------------------
y <-sunflower$YLD
y #Dim (12,1)
```

```
##  [1] 1489 1526 1473 1535 1486 1426 1578 1483 1562 1511 1463 1496
```

3 Fitting a fixed model for environments and years.

```
## Design matrix restricted to environnements-----------------------------------------------------

#Let's create X matrix by hand.
# we use the constraint 'set-first-to-zero'

Xmanual <- matrix(c(1,1,0,0,0,    #μ + Effect of experimental field (i.e Odars) for Measure1
                    1,1,0,0,0,      #μ + Effect of experimental field (i.e Odars) for Measure2
                    1,0,1,0,0,      #μ + Effect of experimental field (i.e. Verfeil) for Measure3
                    1,0,1,0,0,    #μ + Effect of experimental field (i.e. Verfeil) for Measure4
                    1,0,0,1,0,    #μ + Effect of experimental field (i.e. Villefranche)for Measure5
                    1,0,0,1,0,
                    1,0,0,1,0,
                    1,0,0,0,1,
                    1,0,0,0,1,
                    1,0,0,0,1,
                    1,0,0,0,0,
                    1,0,0,0,0),
                  byrow=TRUE,nrow=12,ncol=5)
dimnames(Xmanual)<-list(c("Measure1","Measure2", "Measure3", "Measure4", "Measure5", "Measure6", "Measu
                          c("Intercept", "Odars", "Verfeil", "Villefranche", "Fourquevaux"))
Xmanual
```

```
##           Intercept Odars Verfeil Villefranche Fourquevaux
## Measure1          1     1       0            0           0
## Measure2          1     1       0            0           0
## Measure3          1     0       1            0           0
## Measure4          1     0       1            0           0
## Measure5          1     0       0            1           0
## Measure6          1     0       0            1           0
## Measure7          1     0       0            1           0
## Measure8          1     0       0            0           1
## Measure9          1     0       0            0           1
## Measure10         1     0       0            0           1
## Measure11         1     0       0            0           0
## Measure12         1     0       0            0           0
```

```
#another way to create X is
X <- model.matrix(~sunflower$ENV)
X_env_year <- model.matrix(~sunflower$ENV+sunflower$YEAR)
#Rq: In this dataset, no blocks, no replicates by enviro and year so no possibility to take
# into account the Site x Year interaction.
#If it was the case: X_envXyear <- model.matrix(~sunflower$ENV*sunflower$YEAR)


#2 solutions to solve linear syst
#Sol. 1: ANOVA Constraint : sum to 0: Sum of alpha effects (alpha Env1 + alpha Env2 + alpha Env3=0) but
#solution 2: ANOVA Constraint : set to 0: effect of the 1st modalité by alphabetical order (here, Balma
# Effect of this enviro = intercept.
#The X matrix created with the function contains  (Env-1) columns: Balma column is not present.
# Balma is the first alphabetical Env.
X
```

```
##    (Intercept) sunflower$ENVFourquevaux sunflower$ENVOdars
## 1            1                        0                  1
## 2            1                        0                  1
## 3            1                        0                  0
## 4            1                        0                  0
## 5            1                        0                  0
## 6            1                        0                  0
## 7            1                        0                  0
## 8            1                        1                  0
## 9            1                        1                  0
## 10           1                        1                  0
## 11           1                        0                  0
## 12           1                        0                  0
##    sunflower$ENVVerfeil sunflower$ENVVillefranche
## 1                     0                        0
## 2                     0                        0
## 3                     1                        0
## 4                     1                        0
## 5                     0                        1
## 6                     0                        1
## 7                     0                        1
## 8                     0                        0
## 9                     0                        0
## 10                    0                        0
## 11                    0                        0
## 12                    0                        0
## attr(,"assign")
## [1] 0 1 1 1 1
## attr(,"contrasts")
## attr(,"contrasts")$`sunflower$ENV`
## [1] "contr.treatment"
```

```
#Identical to the X matrix we designed manually except that by default the function model.matrix order
# Factor level by alphabetical order and remove the 1st level

EnvBetas=solve((t(X)%*%X))%*%t(X)%*%y
EnvBetas
```

```
##                                [,1]
## (Intercept)               1479.50000
## sunflower$ENVFourquevaux    39.16667
## sunflower$ENVOdars          28.00000
## sunflower$ENVVerfeil        24.50000
## sunflower$ENVVillefranche   17.16667
```

```
#Effect of Balma = 0 due to constraint "set-to-zero"
#Estimated mean = 1479.50000
Env_yearBetas=solve((t(X_env_year)%*%X_env_year))%*%t(X_env_year)%*%y
Env_yearBetas
```

```
##                                [,1]
## (Intercept)               1534.75
## sunflower$ENVFourquevaux    20.75
## sunflower$ENVOdars          28.00
## sunflower$ENVVerfeil        24.50
## sunflower$ENVVillefranche   -1.25
## sunflower$YEAR2005         -40.15
## sunflower$YEAR2007         -70.35
```

```
#Beta_Balma vaut 0 et Beta_2003 vaut 0
#Balma Estimated mean = intercept and 2003 Estimated mean = intercept.
#If Site X Year interaction in the model then intercept = Estimated mean for Balma_2003.
```

```
#Decomposition of the solving of this linear system
#solve(t(X)%*%X)#Calcule l'inverse de (t(X)%*%X)
#solve(t(X)%*%X)%*%(t(X)%*%X)#  Identity Matrix identité "à l'erreur près" proving what is computed by
#t(X)%*%y
```

4 Fitting a random model to estimate genetic effects.

```
## Modèle aléatoire avec les génotypes en effet aléatoire:
#création de la matrice Z des effets aléatoires et du vecteurs μ des BLUP
sunflower
```

```
##             ENV       Geno YEAR  YLD
## 1         Odars NK Countri 2007 1489
## 2         Odars    PR64H41 2005 1526
## 3       Verfeil NK Countri 2007 1473
## 4       Verfeil    PR64H41 2005 1535
## 5  Villefranche    PR64H41 2007 1486
## 6  Villefranche     Atomic 2005 1426
## 7  Villefranche   Santiago 2003 1578
## 8   Fourquevaux NK Countri 2007 1483
## 9   Fourquevaux      Boogy 2005 1562
## 10  Fourquevaux    Aurasol 2003 1511
## 11        Balma    PR64H41 2007 1463
## 12        Balma    PR64H41 2005 1496
```

```
Zmanual<- matrix(c(1,0,0,0,0,0,   #Ind.1=μ1
                   0,1,0,0,0,0,     #Ind.2=μ2
                   1,0,0,0,0,0,     #Ind.3=μ3
                   0,1,0,0,0,0,
                   0,1,0,0,0,0,
                   0,0,1,0,0,0,
                   0,0,0,1,0,0,
                   1,0,0,0,0,0,
                   0,0,0,0,1,0,
                   0,0,0,0,0,1,
                   0,1,0,0,0,0,
                   0,1,0,0,0,0
),byrow=TRUE,nrow=12,ncol=6)
dimnames(Zmanual)<-list(c("Measure1","Measure2", "Measure3", "Measure4", "Measure5", "Measure6", "Measu
                  c("NK Countri", "PR64H41", "Atomic", "Santiago", "Boogy", "Aurasol"))

Zmanual
```

```
##           NK Countri PR64H41 Atomic Santiago Boogy Aurasol
## Measure1           1       0      0        0     0       0
## Measure2           0       1      0        0     0       0
## Measure3           1       0      0        0     0       0
## Measure4           0       1      0        0     0       0
## Measure5           0       1      0        0     0       0
## Measure6           0       0      1        0     0       0
## Measure7           0       0      0        1     0       0
## Measure8           1       0      0        0     0       0
## Measure9           0       0      0        0     1       0
## Measure10          0       0      0        0     0       1
## Measure11          0       1      0        0     0       0
## Measure12          0       1      0        0     0       0
```

```
Z <- model.matrix(~sunflower$Geno-1)
#By default, model.matrix add an intercept. However, for random effect, no dependencies so no intercept
#Par défaut, Geno classés par ordre alphabétique dans la matrice Z
```

```
Z
```

```
##    sunflower$GenoAtomic sunflower$GenoAurasol sunflower$GenoBoogy
## 1                     0                     0                   0
## 2                     0                     0                   0
## 3                     0                     0                   0
## 4                     0                     0                   0
## 5                     0                     0                   0
## 6                     1                     0                   0
## 7                     0                     0                   0
## 8                     0                     0                   0
## 9                     0                     0                   1
## 10                    0                     1                   0
## 11                    0                     0                   0
## 12                    0                     0                   0
##    sunflower$GenoNK Countri sunflower$GenoPR64H41 sunflower$GenoSantiago
## 1                          1                     0                      0
## 2                          0                     1                      0
## 3                          1                     0                      0
## 4                          0                     1                      0
## 5                          0                     1                      0
## 6                          0                     0                      0
## 7                          0                     0                      1
## 8                          1                     0                      0
## 9                          0                     0                      0
## 10                         0                     0                      0
## 11                         0                     1                      0
## 12                         0                     1                      0
## attr(,"assign")
## [1] 1 1 1 1 1 1
## attr(,"contrasts")
## attr(,"contrasts")$`sunflower$Geno`
## [1] "contr.treatment"
```

```
## Calculation of BLUP from centered values of phenotypes (y -mean(y))
# without taking into account environment effects
# here, we want to fit the following model : Prédiction model for BLUP: y2 = mean(y) + Zu + E
```

Solving the system for random effects needs to take into account the VarCov matrix of the random effects (i.e. that will be later the kinship matrix). In this first analysis, we consider that there is no kinship between genotypes and we will use identity matrix. I = diag(6)

```
y2 <- (sunflower$YLD) - mean(sunflower$YLD) #Obs. Mean = 1502.333
uhatSimple <- solve(t(Z) %*% Z + 1*diag(6)) %*% t(Z) %*% y2
#1*diag(6)=lambdaG with sigmaE/sigmaU arbitrarily set to 1

uhatSimple
```

```
##                                [,1]
## sunflower$GenoAtomic     -38.1666667
## sunflower$GenoAurasol      4.3333333
## sunflower$GenoBoogy       29.8333333
## sunflower$GenoNK Countri -15.5000000
## sunflower$GenoPR64H41     -0.9444444
## sunflower$GenoSantiago    37.8333333
```

5 Fitting a M.E.T. with a mixed model.

MultiEnvironment Trials: Estimation of genetic values but NOT of the breeding values (additive)

```r
# Mixed model: Fixed effect for Environnements & random effects for genotypes

library(lme4)
modellmer<-lmer(sunflower$YLD ~ sunflower$ENV + (1|sunflower$Geno))
#By default, lmer uses G = SigmaU*I & R = SigmaE*I
modellmer
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: sunflower$YLD ~ sunflower$ENV + (1 | sunflower$Geno)
## REML criterion at convergence: 75.9668
## Random effects:
##  Groups         Name        Std.Dev.
##  sunflower$Geno (Intercept) 54.36
##  Residual                   17.96
## Number of obs: 12, groups:  sunflower$Geno, 6
## Fixed Effects:
##               (Intercept)   sunflower$ENVFourquevaux
##                   1477.26                     51.32
##        sunflower$ENVOdars       sunflower$ENVVerfeil
##                     51.13                     47.63
## sunflower$ENVVillefranche
##                      9.50
```

```r
#Compare the BLUE for environnements from this model and the ones computed with fixed effects only:
EnvBetas
```

```
##                                [,1]
## (Intercept)                1479.50000
## sunflower$ENVFourquevaux     39.16667
## sunflower$ENVOdars           28.00000
## sunflower$ENVVerfeil         24.50000
## sunflower$ENVVillefranche    17.16667
```

```r
ranef(modellmer)
```

```
## $`sunflower$Geno`
##             (Intercept)
## Atomic       -54.779240
## Aurasol      -15.844801
## Boogy         30.135958
## NK Countri   -44.014396
## PR64H41        2.241027
## Santiago      82.261451
```

```r
#Compare the BLUP for genotypes from this model and the ones computed with random effects only:
uhatSimple
```

```
##                                [,1]
## sunflower$GenoAtomic     -38.1666667
## sunflower$GenoAurasol      4.3333333
## sunflower$GenoBoogy       29.8333333
## sunflower$GenoNK Countri -15.5000000
## sunflower$GenoPR64H41     -0.9444444
## sunflower$GenoSantiago    37.8333333
```

```r
#To visualize the X design matrix and Z random effect matrix
getME(modellmer,'X')
```

```
##    (Intercept) sunflower$ENVFourquevaux sunflower$ENVOdars
## 1            1                        0                 1
## 2            1                        0                 1
## 3            1                        0                 0
## 4            1                        0                 0
## 5            1                        0                 0
```

```
## 6            1                        0                 0
## 7            1                        0                 0
## 8            1                        1                 0
## 9            1                        1                 0
## 10           1                        1                 0
## 11           1                        0                 0
## 12           1                        0                 0
##    sunflower$ENVVerfeil sunflower$ENVVillefranche
## 1                     0                         0
## 2                     0                         0
## 3                     1                         0
## 4                     1                         0
## 5                     0                         1
## 6                     0                         1
## 7                     0                         1
## 8                     0                         0
## 9                     0                         0
## 10                    0                         0
## 11                    0                         0
## 12                    0                         0
## attr(,"assign")
## [1] 0 1 1 1 1
## attr(,"contrasts")
## attr(,"contrasts")$`sunflower$ENV`
## [1] "contr.treatment"
##
## attr(,"msgScaleX")
## character(0)
```

```
t(as.matrix(getME(modellmer,'Zt')))#unexpected complex syntax...Sorry...
```

```
##    Atomic Aurasol Boogy NK Countri PR64H41 Santiago
## 1       0       0     0          1       0        0
## 2       0       0     0          0       1        0
## 3       0       0     0          1       0        0
## 4       0       0     0          0       1        0
## 5       0       0     0          0       1        0
## 6       1       0     0          0       0        0
## 7       0       0     0          0       0        1
## 8       0       0     0          1       0        0
## 9       0       0     1          0       0        0
## 10      0       1     0          0       0        0
## 11      0       0     0          0       1        0
## 12      0       0     0          0       1        0
```

```
#In "modellmer", Please see:

#Groups     Name        Std.Dev.
# sunflower$Geno (Intercept) 54.36
# Residual                   17.96
#VarG=54.36^2 VarE=17.96^2 h²= Broad Sense heritability
# except if genotypes are pure lines and design is balanced (narrow-sense heritability).
## Caution: If unbalanced design as here, the environnement effects and genotype effects are not
## independent, so this does not provide an accurate Broad sense heritability.


##estimate of broad-sense heritability = square of accuracy
as.data.frame(VarCorr(modellmer))#vcov for heritability calculation
```

```
##              grp        var1 var2      vcov    sdcor
## 1 sunflower$Geno (Intercept) <NA> 2954.6705 54.35688
## 2       Residual        <NA> <NA>  322.5307 17.95914
```

```
# (i.e. VarG & Var E) are in column 4
(varianceestimates<-as.data.frame(VarCorr(modellmer))[,4])
```

```
## [1] 2954.6705  322.5307
```

```
h2hat<-varianceestimates[1]/sum(varianceestimates)
h2hat
```

```
## [1] 0.9015835
```

```
varianceestimates[1]/varianceestimates[2] #SigmaE/SigmaU 9.160897
```

```
## [1] 9.160897
```

6 Estimating genetic effects, taking into account relationships between varieties

```
###
#Reminder: In the previous example, we do not know anything about the relationships between the varieti
#So we used the identity matrix I in the model.
```

6.1 Fitting a random model using pedigrees.

```
## Model P-BLUP (P for pedigree) based on a (fake) pedigree matrix

Amat<- matrix(c(1,0.5,0.25,0,0.25,0,
                0.5,1,0,0.25,0.5,0.25,
                0.25,0,1,0.5,0.5,0,
                0,0.25,0.5,1,0,0.25,
                0.25,0.5,0.5,0,1,0.5,
                0,0.25,0,0.25,0.5,1
               ),byrow=TRUE,nrow=6,ncol=6)
dimnames(Amat)<-list(c("NK Countri", "PR64H41", "Atomic", "Santiago", "Boogy", "Aurasol"),
                     c("NK Countri", "PR64H41", "Atomic", "Santiago", "Boogy", "Aurasol"))
Amat
```

```
##             NK Countri PR64H41 Atomic Santiago Boogy Aurasol
## NK Countri        1.00    0.50   0.25     0.00  0.25    0.00
## PR64H41           0.50    1.00   0.00     0.25  0.50    0.25
## Atomic            0.25    0.00   1.00     0.50  0.50    0.00
## Santiago          0.00    0.25   0.50     1.00  0.00    0.25
## Boogy             0.25    0.50   0.50     0.00  1.00    0.50
## Aurasol           0.00    0.25   0.00     0.25  0.50    1.00
```

```
#Clone: 1; Full sibs: 0.5; Half sibs: 0.25; 1st cousins: 1/8; Parent-offsprings: 0.5; GrandParents/Gran
breedingvalue.withA <- solve(t(Z) %*% Z + 1*solve(Amat)) %*% t(Z) %*% y2
breedingvalue.withA
```

```
##                               [,1]
## sunflower$GenoAtomic     -29.605360
## sunflower$GenoAurasol    -20.319598
## sunflower$GenoBoogy       -2.514109
## sunflower$GenoNK Countri  -2.545691
## sunflower$GenoPR64H41      9.199237
## sunflower$GenoSantiago    21.745537
```

```
#Compare the BLUP for genotypes from this model and the ones computed with random effects only.
uhatSimple
```

```
##                                [,1]
```

```
## sunflower$GenoAtomic     -38.1666667
## sunflower$GenoAurasol      4.3333333
## sunflower$GenoBoogy       29.8333333
## sunflower$GenoNK Countri -15.5000000
## sunflower$GenoPR64H41     -0.9444444
## sunflower$GenoSantiago    37.8333333
```

```
# Compare the BLUP for genotypes from this model and the ones computed with a Mixed model: Fixed effect
ranef(modellmer)
```

```
## $`sunflower$Geno`
##             (Intercept)
## Atomic      -54.779240
## Aurasol     -15.844801
## Boogy        30.135958
## NK Countri  -44.014396
## PR64H41       2.241027
## Santiago     82.261451
```

6.2 Fitting a random model using a kinship matrix.

6.2.1 Marker genotypes and imputation of missing genotypic data.

```
#########
## Taking into account of a Kinship matrix
#FAKE DATA: 6 genotypes and 10 markers - Imputation of genotypes and calculation of the Kinship matrix
#########

MarkersMatrix = read.table("MToImp.csv", sep=";", header=TRUE) #Reads your file
dim(MarkersMatrix)  #Tell you the dimensions of object MarkersMatrix in rows and columns
```

```
## [1]  6 10
```

```
MarkersMatrix
```

```
##   marker.1 marker.2 marker.3 marker.4 marker.5 marker.6 marker.7 marker.8
## 1        1        1        0       -1        1       -1       NA        0
## 2       -1        0       NA       -1        1       NA       -1       NA
## 3        0       NA        1       -1        1       NA        0        1
## 4       -1        1        1       NA       NA        0        1       NA
## 5        0       -1       NA       NA        0       -1        0       -1
## 6        1       -1        0        0       NA       NA        0       -1
##   marker.9 marker.10
## 1        0        NA
## 2       -1         0
## 3       NA        NA
## 4       NA         1
## 5       -1         1
## 6       NA         1
```

```
MForImp= as.matrix(MarkersMatrix) #Creates matrix from imported data
is.matrix(MForImp)
```

```
## [1] TRUE
```

```
dimnames(MForImp)<-list(c("NK Countri", "PR64H41", "Atomic", "Santiago", "Boogy", "Aurasol"),
                        c("M1","M2", "M3", "M4", "M5", "M6", "M7", "M8", "M9", "M10")
                        )
dim(MForImp)
```

```
## [1]  6 10
```

```
MForImp #Coding for alleles depends on their frequency :
```

```
##            M1 M2 M3 M4 M5 M6 M7 M8 M9 M10
## NK Countri  1  1  0 -1  1 -1 NA  0  0  NA
## PR64H41    -1  0 NA -1  1 NA -1 NA -1   0
## Atomic      0 NA  1 -1  1 NA  0  1 NA  NA
## Santiago   -1  1  1 NA NA  0  1 NA NA   1
## Boogy       0 -1 NA NA  0 -1  0 -1 -1   1
## Aurasol     1 -1  0  0 NA NA  0 -1 NA   1
```

```
# minor hmz allele : +1, htz: 0, major hmz allele: -1

library(rrBLUP)
Gimputed <- A.mat(MForImp, max.missing = NULL,
                  impute.method="mean", return.imputed = T)
#For each marker,imputation of the missing data by the mean of th observed genotypes.
Gimputed$imputed #To print the matrix of imputed genotypes.
```

```
##            M1 M2  M3    M4   M5            M6 M7    M8            M9  M10
## NK Countri  1  1 0.0 -1.00 1.00 -1.000000e+00  0  0.00  5.551115e-17 0.75
## PR64H41    -1  0 0.5 -1.00 1.00 -6.666667e-01 -1 -0.25 -1.000000e+00 0.00
## Atomic      0  0 1.0 -1.00 1.00 -6.666667e-01  0  1.00 -6.666667e-01 0.75
## Santiago   -1  1 1.0 -0.75 0.75  5.551115e-17  1 -0.25 -6.666667e-01 1.00
## Boogy       0 -1 0.5 -0.75 0.00 -1.000000e+00  0 -1.00 -1.000000e+00 1.00
## Aurasol     1 -1 0.0  0.00 0.75 -6.666667e-01  0 -1.00 -6.666667e-01 1.00
```

```
# Ex: marker 2: 3*1/4=0.75; marker3 (2*-1+1)4=-0.25

#Another possible method to impute the missing genotype: EM algorithm
Gimputed.EM <- A.mat(MForImp, max.missing = NULL,
                     impute.method="EM", return.imputed = T)
```

```
## [1] "A.mat converging:"
## [1] 0.11
## [1] 0.0222
## [1] 0.00899
```

```
Gimputed.EM$imputed
```

```
##            M1            M2        M3        M4         M5            M6
## NK Countri  1  1.000000e+00 0.0000000 -1.000000 1.00000000 -1.000000e+00
## PR64H41    -1  0.000000e+00 0.8942998 -1.000000 1.00000000 -6.415691e-01
## Atomic      0  3.330669e-16 1.0000000 -1.000000 1.00000000 -2.982424e-01
## Santiago   -1  1.000000e+00 1.0000000 -0.833208 1.41681022  5.551115e-17
## Boogy       0 -1.000000e+00 0.1057002 -0.666792 0.00000000 -1.000000e+00
## Aurasol     1 -1.000000e+00 0.0000000  0.000000 0.08318978 -1.060188e+00
##                        M7          M8            M9       M10
## NK Countri -8.881784e-16  0.00000000  5.551115e-17 0.5079329
## PR64H41    -1.000000e+00 -0.05012839 -1.000000e+00 0.0000000
## Atomic      0.000000e+00  1.00000000 -6.006184e-01 0.9920671
## Santiago    1.000000e+00 -0.44987161 -9.186012e-01 1.0000000
## Boogy       0.000000e+00 -1.00000000 -1.000000e+00 1.0000000
## Aurasol     0.000000e+00 -1.00000000 -4.807804e-01 1.0000000
```

```
# rrBLUP::Amat calcule aussi une kinship matrix.
# Attention : cette matrice de kinship dans $A peut présenter des valeurs négatives.
# c'est l'estimateur 'first G' de VanRaden (2008)
# negative genomic relationships : This implies that individuals i and j are
# more distinct than an average pair of individuals in the data
# Fixing negative estimates of relationships to 0 is **wrong** praxis.
```

6.2.2 Computing the kinship matrix "by hand".

```
#Canonical calculation of the Kinship matrix
p <- colSums((Gimputed$imputed + 1)/ 12)  # 6 diploid accessions thus a maximum of 12 alleles
p2 <- 2 * sum(p*(1-p))
Gmatrix <- crossprod(t(Gimputed$imputed))/p2
# this matrix is usally different form that computed with rrBLUP::Amat
```

6.2.3 Prediction of breeding values.

```
# this is a random-only model (w/o fixed environmental effects)
breedingvalue.withG <- solve(t(Z) %*% Z + 1*solve(Gmatrix)) %*% t(Z) %*% y2  # by hand
breedingvalue.withG
```

```
##                                [,1]
## sunflower$GenoAtomic     -25.379928
## sunflower$GenoAurasol      3.946911
## sunflower$GenoBoogy       14.426679
## sunflower$GenoNK Countri -18.098704
## sunflower$GenoPR64H41      7.086380
## sunflower$GenoSantiago    21.105226
```

```
# compare with uhatSimple (same model BUT identity matrix as G) or to
# breedingvalue.withA (same model BUT (fake) pedigree matrix as G)


# this is the mixed model (fixed environments effects and random genetic values)
#library(rrBLUP)
GBLUPmodel <- mixed.solve(y = y,X = X,Z = Z, K = Gmatrix, method = "REML")  # using rrBLUP::mixed.solve
#The function mixed.solve also estimate sigmaU and sigmaE
#G=K*SigmaU and R=I*SigmaE
GBLUPmodel
```

```
## $Vu
## [1] 4115.974
##
## $Ve
## [1] 560.2396
##
## $beta
##               (Intercept)  sunflower$ENVFourquevaux
##                 1480.98716                  71.54115
##        sunflower$ENVOdars      sunflower$ENVVerfeil
##                  57.19312                  53.69312
## sunflower$ENVVillefranche
##                  13.35379
##
## $u
## NK Countri    PR64H41     Atomic   Santiago      Boogy    Aurasol
## -53.905322 -40.091234  -1.620328 -59.873396  -1.487165  62.369615
##
## $LL
## [1] -36.69172
```

```
GBLUPmodel$Vu / (GBLUPmodel$Vu + GBLUPmodel$Ve)
```

```
## [1] 0.8801938
```

```
# Restricted maximum likelihood (REML) because the G matrix is unknown but estimated.
# For manual solving of the Henderson's equation, we use least squares method
# which only apply if we exactly know the G matrix (assumption emitted in this case)
```

```
# with rrBLUP impossible to code for least squares method
# LL : Lod likelihood value : when chosing between 2 models, take the one with the highest LL.
```

7 Predicting phenotypes, only knowing genetic relationships.

Prediction of the phenotype of a new line (which was not phenotyped) on the basis of its genotypic kinship with varieties that have been evaluated in the field

```
###

set.seed(1)
# generating genotypes for three new lines
newline1 <- sample(c(-1,0,1),10,replace=T)
newline2 <- sample(c(-1,0,1),10,replace=T)
newline3 <- sample(c(-1,0,1),10,replace=T)
Mnew <- rbind(Gimputed$imputed,newline1,newline2,newline3)
rownames(Mnew) <- c("Atomic",
                    "Santiago",
                    "NK Countri",
                    "Boogy",
                    "Aurasol",
                    "PR64H41",
                    "newline1",
                    "newline2",
                    "newline3")
Mnew
```

```
##            M1 M2  M3    M4    M5            M6 M7    M8            M9
## Atomic      1  1 0.0 -1.00  1.00 -1.000000e+00  0  0.00  5.551115e-17
## Santiago   -1  0 0.5 -1.00  1.00 -6.666667e-01 -1 -0.25 -1.000000e+00
## NK Countri  0  0 1.0 -1.00  1.00 -6.666667e-01  0  1.00 -6.666667e-01
## Boogy      -1  1 1.0 -0.75  0.75  5.551115e-17  1 -0.25 -6.666667e-01
## Aurasol     0 -1 0.5 -0.75  0.00 -1.000000e+00  0 -1.00 -1.000000e+00
## PR64H41     1 -1 0.0  0.00  0.75 -6.666667e-01  0 -1.00 -6.666667e-01
## newline1   -1  0 0.0  1.00 -1.00  1.000000e+00  1  0.00  0.000000e+00
## newline2   -1 -1 1.0  0.00  1.00  0.000000e+00  1  1.00  0.000000e+00
## newline3    1 -1 0.0 -1.00 -1.00  0.000000e+00 -1  0.00  1.000000e+00
##              M10
## Atomic      0.75
## Santiago    0.00
## NK Countri  0.75
## Boogy       1.00
## Aurasol     1.00
## PR64H41     1.00
## newline1   -1.00
## newline2    1.00
## newline3    0.00
```

```
## Calculation of the kinship matrix: relationships of the new lines with the first set of accessions
```

```
Mnew+1
```

```
##            M1 M2  M3   M4   M5        M6 M7   M8        M9  M10
## Atomic      2  2 1.0 0.00 2.00 0.0000000  1 1.00 1.0000000 1.75
## Santiago    0  1 1.5 0.00 2.00 0.3333333  0 0.75 0.0000000 1.00
## NK Countri  1  1 2.0 0.00 2.00 0.3333333  1 2.00 0.3333333 1.75
## Boogy       0  2 2.0 0.25 1.75 1.0000000  2 0.75 0.3333333 2.00
## Aurasol     1  0 1.5 0.25 1.00 0.0000000  1 0.00 0.0000000 2.00
## PR64H41     2  0 1.0 1.00 1.75 0.3333333  1 0.00 0.3333333 2.00
## newline1    0  1 1.0 2.00 0.00 2.0000000  2 1.00 1.0000000 0.00
## newline2    0  0 2.0 1.00 2.00 1.0000000  2 2.00 1.0000000 2.00
```

```
## newline3    2  0 1.0 0.00 0.00 1.0000000  0 1.00 2.0000000 1.00
```

```
p <-colSums(Mnew+1)/ 18  # now 9 diploid genotypes, so 18 alleles
sum(2*p*(1-p))
```

```
## [1] 4.425926
```

```
Gnew <- Mnew %*% t(Mnew)/ (sum(2*p*(1-p)))   ## Van Raden's 'first G'
Gnew
```

```
##                  Atomic   Santiago NK Countri       Boogy    Aurasol
## Atomic      1.256799e+00  0.3765690  0.7296025  0.50836820  0.3389121
## Santiago    3.765690e-01  1.3007322  0.7594142  0.61663180  0.6589958
## NK Countri  7.296025e-01  0.7594142  1.2316946  0.77824268  0.5271967
## Boogy       5.083682e-01  0.6166318  0.7782427  1.49843096  0.4471757
## Aurasol     3.389121e-01  0.6589958  0.5271967  0.44717573  1.3132845
## PR64H41     4.895397e-01  0.2510460  0.3138075  0.05805439  0.9790795
## newline1   -1.073222e+00 -0.6025105 -0.7719665 -0.11297071 -0.6213389
## newline2   -5.648536e-02  0.2824268  0.8472803  0.79079498  0.3389121
## newline3    1.254227e-17 -0.2259414 -0.1506276 -0.82845188  0.1694561
##                PR64H41   newline1    newline2      newline3
## Atomic      0.48953975 -1.0732218 -0.05648536  1.254227e-17
## Santiago    0.25104603 -0.6025105  0.28242678 -2.259414e-01
## NK Countri  0.31380753 -0.7719665  0.84728033 -1.506276e-01
## Boogy       0.05805439 -0.1129707  0.79079498 -8.284519e-01
## Aurasol     0.97907950 -0.6213389  0.33891213  1.694561e-01
## PR64H41     1.23169456 -0.7719665  0.16945607  1.317992e-01
## newline1   -0.77196653  1.3556485  0.00000000 -4.518828e-01
## newline2    0.16945607  0.0000000  1.58158996 -4.518828e-01
## newline3    0.13179916 -0.4518828 -0.45188285  1.355649e+00
```

```
dim(Gnew) #  pay attention to the dimensions.
```

```
## [1] 9 9
```

```
(sunflower2 <- sunflower)
```

```
##             ENV       Geno YEAR  YLD
## 1         Odars NK Countri 2007 1489
## 2         Odars    PR64H41 2005 1526
## 3       Verfeil NK Countri 2007 1473
## 4       Verfeil    PR64H41 2005 1535
## 5  Villefranche    PR64H41 2007 1486
## 6  Villefranche     Atomic 2005 1426
## 7  Villefranche   Santiago 2003 1578
## 8   Fourquevaux NK Countri 2007 1483
## 9   Fourquevaux      Boogy 2005 1562
## 10  Fourquevaux    Aurasol 2003 1511
## 11        Balma    PR64H41 2007 1463
## 12        Balma    PR64H41 2005 1496
```

```
sunflower2$Geno <- factor(as.character(sunflower$Geno),
                          levels=rownames(Gnew))
Znew <- model.matrix(~ sunflower2$Geno-1)
dim(Znew)
```

```
## [1] 12  9
```

```
## THIS dimension IS VERY important : 12 realized observations and 9 genetic effects to estimate
# 6 genotypes with phenotypes and 3 genotypes W/O phenotypes.


## Prediction using a random-only model: Caution ! we do not take into account the effect of the enviro
```

```r
breedingvalue.new <- solve(t(Znew) %*% Znew + 1 * solve(Gnew)) %*% t(Znew) %*% y2
# here : SigmaE/SigmaU = 1 by hypothesis
breedingvalue.new
```

```
##                                [,1]
## sunflower2$GenoAtomic     -35.896593
## sunflower2$GenoSantiago    33.967595
## sunflower2$GenoNK Countri -10.198162
## sunflower2$GenoBoogy       26.915408
## sunflower2$GenoAurasol     17.329729
## sunflower2$GenoPR64H41     -4.573932
## sunflower2$Genonewline1    30.920099
## sunflower2$Genonewline2    13.330409
## sunflower2$Genonewline3   -31.024520
```

```
## prediction using the full mixed model.
```

```r
mixed.solve(y = y, X = X, Z = Znew, K = Gnew, method = "REML")  # mixed.solve estimates SigmaU and Sigm
```

```
## $Vu
## [1] 5180.895
##
## $Ve
## [1] 391.3955
##
## $beta
##               (Intercept)  sunflower$ENVFourquevaux
##                1504.93453                 37.00634
##        sunflower$ENVOdars      sunflower$ENVVerfeil
##                  46.81971                 43.31971
## sunflower$ENVVillefranche
##                  10.35674
##
## $u
##     Atomic   Santiago NK Countri      Boogy    Aurasol    PR64H41
##  -83.05699   52.61770  -63.07395   14.85885  -21.60752  -25.43453
##   newline1   newline2   newline3
##   74.24884  -13.76352  -78.77167
##
## $LL
## [1] -36.2682
```